LEARNER GUIDE

Putting it together — scale plans

Constructions, Loci and Bearings · Lesson 7. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

A plan is a shrunk copy of a real place, drawn to a fixed **scale** such as 1 cm : 5 m (every 1 cm on the plan stands for 5 m). That one ratio lets you turn a plan length into a real distance and back again. In this final lesson you bring the whole unit together: use a scale, pick the right construction for each rule (bisector, circle, distance from a line), and shade the **region** that satisfies every rule at once.

What you are learning to notice

- A **scale** links plan and reality: plan → real means **multiply**; real → plan means **divide**.
- Each rule about position is a **locus** you already know — a circle, a bisector, or a distance from a line.
- The answer to a compound problem is the **region where the loci overlap**.
- Choosing the right construction for each rule is the heart of a plan problem.

Moving through it, one step at a time

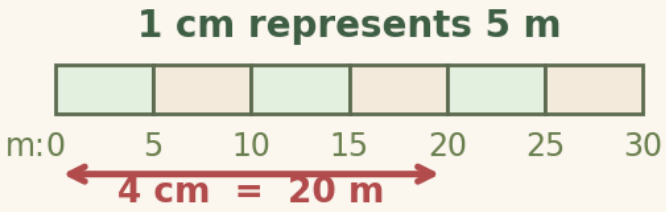
First fix the scale and convert every real distance into a plan length (divide by the scale). Then, for each rule, draw its boundary with the right construction: a circle for 'a fixed distance from a point'; the perpendicular bisector for 'equidistant from two points'; the angle bisector for 'equidistant from two lines'; a parallel line for 'a fixed distance from a fence'. Finally, shade the region that obeys every rule at once — where the boundaries overlap. Leave your construction marks showing.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Converting with a scale

A plan is drawn at 1 cm : 5 m. A path measures 4 cm on the plan. How long is it really?

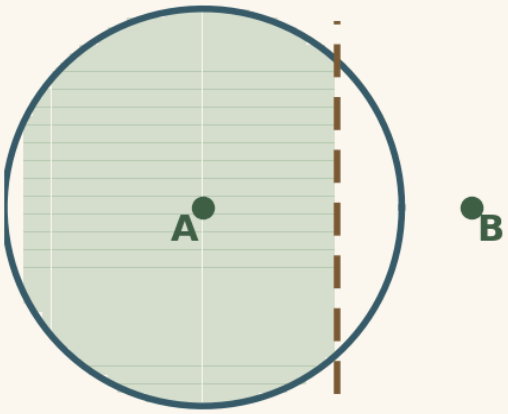


1. The scale is 1 cm represents 5 m.	1 cm : 5 m
2. Plan to real: multiply the plan length by 5.	4 × 5
3. So the real length is 20 m.	= 20 m

20 m — a plan length of 4 cm at 1 cm : 5 m is 4 × 5 = 20 m. (To go back, divide by 5.)

Example 2 — A region from two rules

On a plan, shade where a feature is within a distance of A AND closer to A than to B.



1. 'Within a distance of A' — draw a circle centred A; inside obeys the rule.	circle at A
2. 'Closer to A than B' — construct the perpendicular bisector of AB; A's side obeys it.	perp. bisector
3. Shade where both are true — inside the circle and on A's side.	overlap

The region is the part of the circle on A's side of the perpendicular bisector — where the two loci overlap.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the whole unit — bisectors, loci and regions, and bearings), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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